1 Preliminaries

Definition 1. A sequence (a_n) of real numbers is bounded if there is $M > 0$ such that $|a_n| \le M$ for all n.

Definition 2. A subsequence of (a_n) is a sequence (a_{n_k}) where $n_1 < n_2 < \cdots$ are positive integers.

Remark 1. Boundedness is used below only through the bound M.

Lemma 1. Every sequence of real numbers has a monotone subsequence.

Proof. Call an index n a peak if $a_n \ge a_m$ for all $m > n$. If there are infinitely many peaks, list them as $n_1 < n_2 < \cdots$; then $a_{n_1} \ge a_{n_2} \ge \cdots$ is a monotone subsequence. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Otherwise there is a last peak N, and starting from $m_1 = N + 1$ we can choose indices

$m_1 < m_2 < \cdots$ with

$$a_{m_1} < a_{m_2} < a_{m_3} < \cdots$$

which is again monotone. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. Observe that every index either is a peak or is not a peak, and that the set of indices after a given index is again infinite, so the argument can be repeated without loss of generality. This proves the lemma.

Theorem 2 (Bolzano–Weierstrass). Every bounded sequence of real numbers has a convergent subsequence.

Proof. Let (a_n) be bounded. By Lemma 1 it has a monotone subsequence. Since the sequence is bounded, we may choose a convergent subsequence. Indeed, a bounded monotone sequence converges, and so the subsequence converges to some L with $|L| \leq M$.

Example 1. The sequence $a_n = (-1)^n$ is bounded and has the convergent subsequence $a_{2k} = 1$.

Exercise 1. Show that the hypothesis

of boundedness cannot be dropped.

Notation 1. We write (a_{n_k}) for a subsequence.